

SHERAZ TARIQ

AI/ML Engineer | Generative AI, Agentic AI & RAG | Full Stack AI Developer

☎ +92 340 543 7523 ✉ sheraztariq2978@gmail.com [in linkedin.com/in/sheraz-tariq](https://www.linkedin.com/in/sheraz-tariq) github.com/sheraztariq22 🌐 [Portfolio](#)

Summary

AI/ML Engineer specializing in Generative AI, Agentic (Multi-Agent) Systems, and production-grade RAG pipelines. I turn LLMs from chat experiments into scalable, enterprise-grade products, architecting the robust backends that power them and the full-stack applications that make them accessible.

Experience

SkilliHire

Feb 2026 – Present

Lead Full Stack AI Engineer

Remote – Australia

- Architected and lead development of **NextGen Schooling** (ngedu.ai), an AI-powered homeschooling platform delivering the Australian Curriculum from **Year 0 to Year 12** to students, parents, and facilitators.
- Designed and shipped multiple **LLM agents** – AI tutor, classroom agent, daily coach, and parent advisor – plus real-time **voice tutoring** using AWS Nova Sonic streaming with barge-in and Amazon Polly TTS.
- Built the full-stack product (React, TypeScript, Tailwind, Supabase Postgres/pgvector) with **70+ serverless edge functions**, Stripe subscription billing, and role-based portals for students, parents, facilitators, and admins.
- Owned production concerns end-to-end: AWS infrastructure (Bedrock, S3, Lambda), performance budgets, SEO/GEO optimization, and end-to-end & load testing.

SageTeck

June 2025 – Present

Software Engineer

Islamabad, Pakistan

- Engineered and maintained critical features for enterprise CRM platforms **SugarCRM** and **SuiteCRM**, ensuring high performance and system reliability.
- Developed a **LangGraph**-based multi-agent assistant (“talkerCRM”) that integrates Generative AI into CRM workflows.
- Contributed across the software development lifecycle with a focus on scalable backend development and clean architecture.

Projects

NextGen Education RAG Backend | *FastAPI, AWS Bedrock, Supabase pgvector, Hybrid RAG*

2026

- Engineered the FastAPI RAG microservice powering NextGen Schooling: syllabi are embedded with **Amazon Titan V2** into Supabase **pgvector**, and **Claude 3.5 Sonnet** generates grade-aligned readings, quizzes, and slides on demand.
- Implemented **hybrid retrieval** (BM25 sparse + dense vectors fused via **Reciprocal Rank Fusion**) with reranking, Haiku-driven query expansion, and a semantic query cache to maximize recall and answer relevance.

DocChat – Multi-Agent RAG System | *LangGraph, LangChain, Hybrid Retrieval*

2025

- Built a multi-agent document QA system that validates whether documents can answer a query, generates grounded responses, and cross-checks them against source content to **detect hallucinations**.

Community Automation System | *Blockchain, AI, MERN Stack* (Final Year Project)

2024 – 2025

- Developed a blockchain- and AI-powered community platform enabling transparent, automated community operations with advanced user-interaction features.

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL

GenAI & Agents: LangChain, LangGraph, CrewAI, RAG, Multi-Agent Systems, LLMs (Claude, Gemini), Prompt Engineering

AI/ML: TensorFlow, Keras, Deep Learning, NLP, Computer Vision

Backend & Web: FastAPI, Django, React, Node.js, REST APIs, Streamlit

Cloud & Data: AWS (Bedrock, S3, Lambda, Titan, Nova Sonic), Supabase/pgvector, Vector Search, PostgreSQL, Docker, Kubernetes, CI/CD

Tools: Git, GitHub, Stripe, SugarCRM, SuiteCRM, Jupyter Notebook

Education

National University of Computer & Emerging Sciences (FAST NUCES)

Sep 2021 – June 2025

Bachelor of Science in Software Engineering

Islamabad, Pakistan

Certifications

Generative AI Engineering with LLMs · Deep Learning Specialization · Machine Learning Specialization · Python Data Structures · Version Control